

Name

Class

Date

Scientific Method review 2025

Total questions: 36

Worksheet time: 29mins

1. Why is it advisable to use tongs when taking a beaker off a laboratory burner?
 - a) It prevents spilling.
 - b) It looks cool.
 - c) It prevents the beaker from breaking.
 - d) It allows you to move it if it is too hot.
2. What is the part of an experiment that serves as a standard for comparison?
 - a) Control Group
 - b) Hypothesis
 - c) Constant
 - d) Variable
3. A testable prediction that predicts a cause and effect relationship between variables is a(n)
 - a) analysis
 - b) observation
 - c) hypothesis
 - d) inference
4. Where does the dependent variable go on a graph?
 - a) X-axis
 - b) In the Title
 - c) Y-axis
 - d) In the Key
5. A factor in an experiment that changes in response to the independent variable is the _____.
 - a) constant
 - b) independent variable
 - c) control group
 - d) dependent variable
6. The “cause” of the experiment, what we are purposefully manipulating or changing is the
 - a) constant
 - b) control group
 - c) dependent variable
 - d) independent variable

7. Which of the following is a quantitative observation?
- a) The dog has two legs.
 - b) The dog is small.
 - c) The dog is cute.
 - d) The dog is fuzzy.
8. If your lab partner's clothes catch on fire, you should first _____.
- a) use a fire extinguisher
 - b) do nothing. It will burn out eventually.
 - c) panic
 - d) tell your partner to stop drop and roll
9. All of the following are steps in the scientific method EXCEPT
- a) observing and recording data.
 - b) forming a hypothesis.
 - c) developing a model.
 - d) discarding data that doesn't match the hypothesis.
10. The best type of graph to use to show the percentages of a whole is a
- a) Line Graph
 - b) Circle Graph
 - c) Scatter Graph
11. Mrs. James wants to measure the volume of a liquid very precisely. What is the BEST lab equipment she could use to do this?
- a) beaker
 - b) test tube
 - c) graduated cylinder
 - d) Erlenmeyer flask
12. An experiment was designed to investigate the effect of caffeine on the heartbeat of puppies. Two litters of black Labrador puppies were used. These puppies were the same age, gender, and size. Litter A was given caffeine and Litter B was not. The heartbeat of the puppies was measured every hour for five hours. What is the independent variable in this experiment?
- a) caffeine or not
 - b) heart beat
 - c) litter B
 - d) litter A

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17. Mr. Smith is conducting an experiment to determine the effect of different light colors on the growth of plants. He uses three types of light: red, blue, and green. He measures the height of the plants every day for a month. What is the independent variable in this experiment?

- | | |
|-------------------------|------------------|
| a) height of the plants | b) type of light |
| c) time | d) type of plant |

18. In an experiment, what is the term for the factor that is measured or observed?

- a) constant
- b) control group
- c) dependent variable
- d) independent variable

19. In an experiment, the variable that is deliberately changed by the scientist is the ____.

- a) dependent variable
- b) independent variable
- c) control group
- d) constant

20. A logical procedure for answering a scientific question is known as

- a) the scientific method
- b) theory
- c) a law
- d) a worldview

21. Correctly match the steps of scientific investigation with the correct description.

Carry out testing/trials, that will help
produce evidence to support the
hypothesis.



○ Observation

Specific question ○

○ Problem

A prediction of what the scientists
thinks will happen that can be tested. ○

○ Hypothesis

Examine the information collected and
summarize the the experiment
revealing whether or not the data
found supports their original
hypothesis (what actually happened).



○ Experiment

Using your senses to gather
information and make inferences
(educated conclusion based on
evidence/reason) ○

○ Results/Conclusion

22. The higher the temperature of the air in an oven, the faster a cake bakes.

Time it takes to bake a cake ○

○ Control Variable

Temperature of oven ○

○ Dependent Variable

Type of oven ○

○ Independent Variable

23. Homer notices that his shower is covered in a strange green slime. His friend Barney tells him that coconut juice will get rid of the green slime. Homer decides to check this out by spraying half of the shower with coconut juice. He sprays the other half of the shower with water. After 3 days of "treatment" there is no change in the appearance of the green slime on either side of the shower.

Control Group (a) _____

Independent Variable (b) _____

Dependent Variable (c) _____

Choose from the below words

Shower sprayed with water

Types of shower spray

Presence of green slime

Shower sprayed with coconut juice

24. The experimental factor that is manipulated; what **change** in the experiment. You should only have 1!

a) independent variable

b) problem

c) constant

d) research

25. parts of the experiment that are **kept the same**.

a) constant

b) research

c) results

d) hypothesis

26. Maggie loves fresh brownies. Her grandma told her if she placed a piece of bread in the storage container with the brownies, the brownies will stay fresh and moist longer. Maggie decides to test this idea. She bakes a batch of brownies. She places 6 brownies in 6 individual plastic containers. Each brownie is the same size and placed in the same location. She adds a piece of bread to three of the bread containers and leaves the other three alone. Each day for 3 weeks she measures the moistness of the brownies and records the results. What is the control group?

a) The brownies that were stored with a piece of bread

b) The brownies that were stored without a piece of bread

c) The size of the brownies

d) The type of container used for storage

27. The Y axis is _____?

- a) Vertical
- b) Horizontal
- c) criss-cross
- d) diagonal

28. What would be the best scale for this data?

Days	Cm
1	1.0
2	3.5
3	5.0
4	8.5

- a) count by fives to 10
- b) count by tens to 100
- c) count by ones to 5
- d) count by ones to 10

29. Which variable should be on the y axis?

- a) the dependent variable
- b) the independent variable
- c) the control variable

30. On what axis should the independent variable always be?

- a) X
- b) Y

31. Which type of graph is best for seeing changes over time?

- a) scatter plot
- b) line graph
- c) bar graph
- d) pie chart

32. In an experiment, a scientist is looking at how daily water affects bonsai growth. What are the following components of the experiment? Match the components with the correct vocabulary word.

Daily water ☉

☉ Constant

Light given to bonsai ☉

☉ Dependent Variable

Bonsai growth ☉

☉ Independent Variable

33. **SpongeBob Square pants wanted to see which brand of bubble gum blew the biggest bubble. He chose three brands to test: Bubble Yum, Big Chew, and Tasty Burst.**
- A) Brand of gum**
B) Person blowing the gum
C) Size of the bubble

Name the constant that SpongeBob kept the same. ☐

☐ **B) Person blowing the gum**

Name the independent variable. ☐

☐ **A) Brand of gum**

Name the dependent variable. ☐

☐ **C) Size of the bubble**

34. Measuring the number of times a person breathes after exercise involves ____ data.

a) quantitative

b) qualitative

35. In your own words, describe the steps of the scientific method. Explain why scientists follow these steps and how each step helps make experiments more reliable. Be sure to include all the steps in order and give at least one example of how they connect to each other.

36. *Scientists often use graphs to organize and show data. Explain why graphs are important in science. In your answer, describe how to correctly set up a graph, including labeling the axes, giving it a title, and using the correct units. Give an example of how a graph can help make data easier to understand.*
